SQL commands

1. Data Definition Language (DDL)
2. Data Query Language (DQL/DRL)
3. Data Manipulation Language (DML)
4. Data Control Language (DCL)
5. Transaction control Language (TCL)

Data Definition Language

Why we need to define data - Is it completely m/c controlled or developer controlled - whether database developer is going to define the complete database schema or is it done by system itself?

Careful about DROP → as its permanently deleted.

Customer

ID	F_Name	L_Name

→ CREATE

→ ALTER

→ DROP → Delete all rows of table & also whatever it has in memory

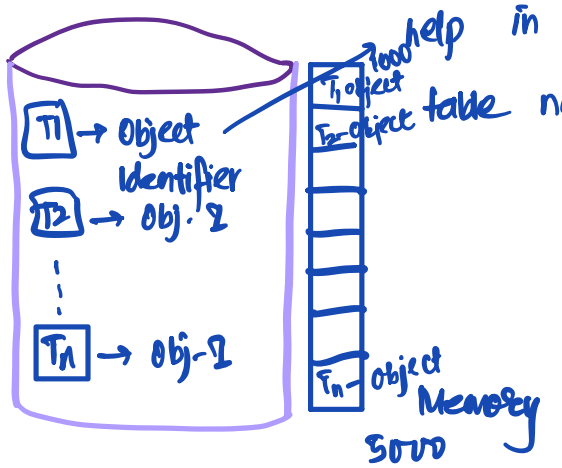
→ TRUNCATE → Just to delete rows from table & not complete table

→ RENAME → Table/DB Name etc

CHAR → VARCHAR

All constraints associated with table can also be deleted using DROP.

MySQL (RDBMS)



help in identifying what is the table name

Data Retrieval / Query Language (mainly used)

→ SELECT FROM

Data manipulation language

→ Transaction Table

TID	Tamt	Tdate	Tdes
1	5000	2010-9-22	Fdk

→ INSERT → new row

→ UPDATE

→ DELETE

changing it

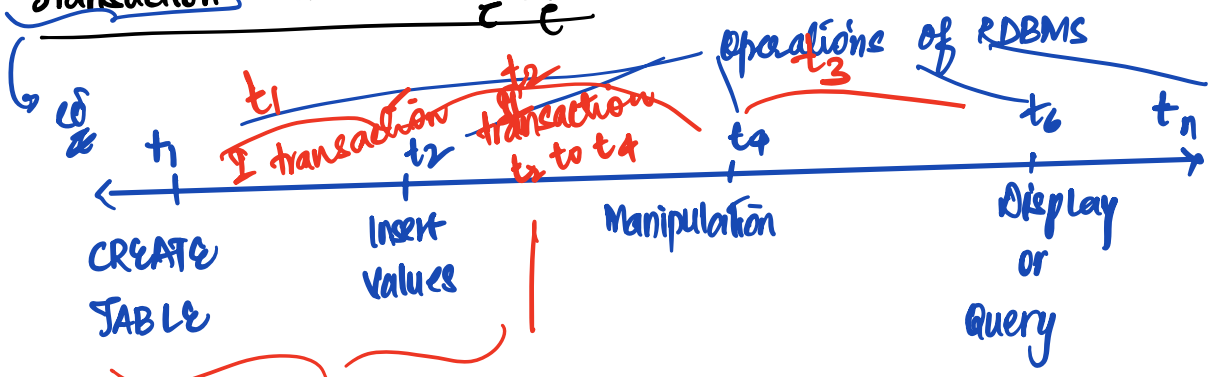
₹50 lakh cr.

eg 10M records

all these can be deleted using DELETE

Thus, before using this, permission is taken, data can be retrieved after its deleted

Transaction Control Language



t_1 to t_n → Operation or work or unit of work

Server may go down — Expectation : All data stored in memory.

lets say till this time - everything is saved in memory

START TRANSACTION; (t_1)

CREATE

INSERT (t_2) → Server keeps on getting down

COMMIT → (helps us saves all info. till this t_2)

Manipulation (t_4) → COMMIT → But everytime using this COMMIT command in b/w operations is not possible

: (t_5)

Display (t_6)

for that, we have <u>Set Commit</u>

SET COMMIT = 0 { Thus no auto-commit
thus, we always need to use COMMIT }

SET COMMIT = 1 → (Everything automatically saved)

<u>SAVE POINT</u> → whenever we want to set up a breakover say t_4 to t_6 — need to put one pass here so that we save pass with <u>save point</u>.

<u>SAVE POINT <name>;</u>

START TRANSACTION;

CREATE (t_1)

INSERT (t_2)

MANIPULATION (t_4)

SAVE POINT t_4 - operation - till;

DISPLAY (t_6)

ROLL BACK t_4 - operations till;
↳ can completely get from t_4 to t_4

Data Query Language

- SELECT
- FROM

Syntax :

SELECT < set of table column_names >

FROM < table - name > ;

Column 1,

Column 2,

⇓

SELECT Column 1, colum2 FROM Customer ;

order of execution :→ from ⟶ SELECT

- we can use select w/o from
  - ↳ Dual Tables (already available in MySQL)
    - ↳ current time of server

eg SELECT 100 + 1 ;

O/P → 101

SELECT ucase ('ak')

O/P AK

SELECT NOW ();

Timestamp

SELECT current_timestamp (5);

one more feature

seconds

upto 5 microseconds

Data control language

→ To grant & take back authority from users

↳ GRANT: Access privileges to the database

↳ REVOKE: withdraw users access privilege.